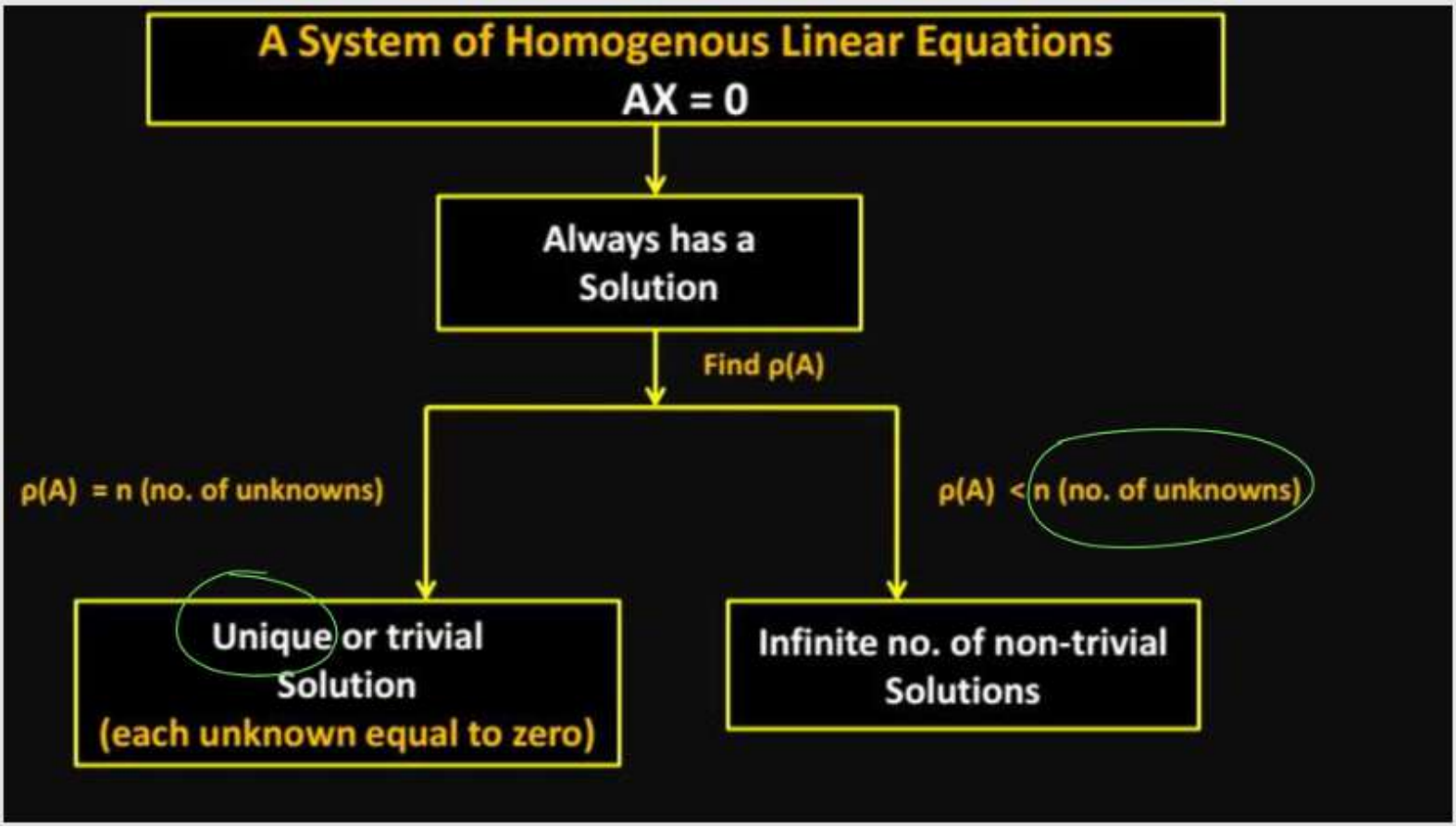
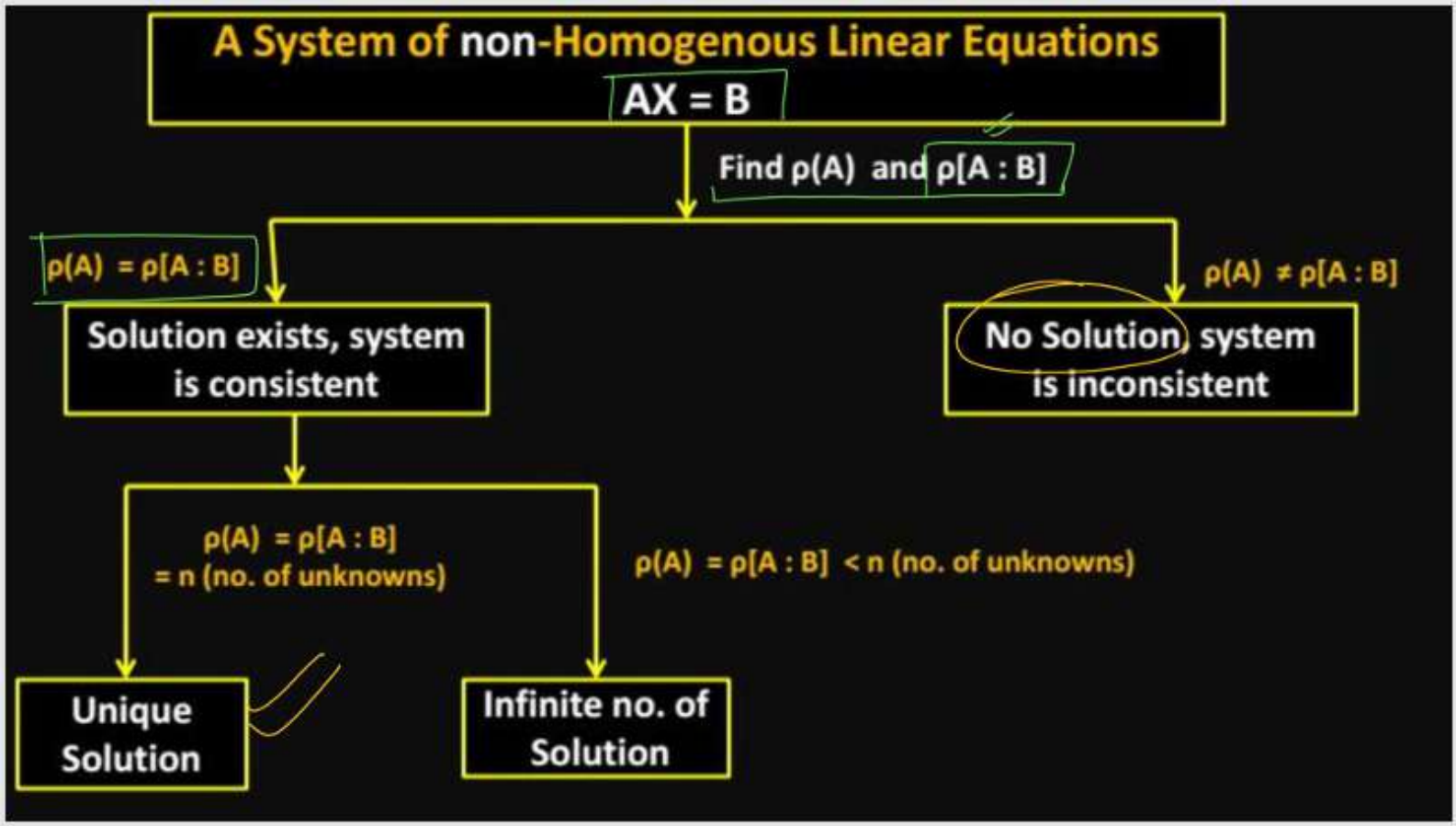

A System of Homogenous Linear Equations
AX = 0
Always has a Solution
Find ρ(A)
ρ(A) = n (no. of unknowns)
ρ(A) < n (no. of unknowns)
Unique or trivial Solution
(each unknown equal to zero)
Infinite no. of non-trivial Solutions


A System of non-Homogenous Linear Equations
AX = B
Find ρ(A) and ρ[A : B]
ρ(A) = ρ[A : B]
Solution exists, system is consistent
ρ(A) ≠ ρ[A : B]
No Solution, system is inconsistent
ρ(A) = ρ[A : B] = n (no. of unknowns)
ρ(A) = ρ[A : B] < n (no. of unknowns)
Unique Solution
Infinite no. of Solution

Practice Q 7.2. Determine the values of λ and μ such that the equations

$$2x - 5y + 2z = 8, \quad 2x + 4y + 6z = 5, \quad x + 2y + \lambda z = \mu$$

have (i) no solution (ii) a unique solution (iii) infinite solutions

Ans. (i) $\lambda=3$ and $\mu\neq5/2$ (ii) $\lambda \neq 3$ and μ may have any value (iii) $\lambda=3$ and $\mu=5/2$

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DPP- 11

Topic : *Regression Lines*

Q.1 *From the data given, find the equation of lines of regression of x on y and y on x. Also, calculate the correlation co − efficient.*

x:	2	4	6	8	10
y:	5	7	9	8	11

Q.2 *For two random variables, x and y with the same mean, the two regression equations are $y = ax + b$ and $x = \alpha y + \beta$. Show that $\frac{b}{\beta} = \frac{1-a}{1-\alpha}$. Find also the common mean.* (AKTU – 2017)

$\bar{x} = \bar{y} = m$

Q.3 *The following data regarding the heights (y) and weights (x) of 100 college students are given:* (AKTU – 2018)

$$\sum x = 15000, \quad \sum x^2 = 2272500, \quad \sum y = 6800, \quad \sum y^2 = 463025 \text{ and } \sum xy = 1022250$$

Find the equation of regression line of height on weight.

Q.4 *The equations of two regression lines, obtained in a correlation analysis of 60 observations are:*

$$5x = 6y + 24 \text{ and } 1000y = 768x - 3608.$$

What is the correlation coefficient? Show that the ratio of coefficient of variablity of x to that of y is $\frac{5}{24}$. What is the ratio of x and y. (AKTU-2019)

Regression Analysis

- Regression means functional relationship between two or more related variables.
- If two varibales are correlated then there exists a relation between them.
- Regression Analysis is a process of estimating relationship between dependent variable and independent variables.

Linear Regression

When points in scatter diagram are concentrated around a straight line then it is Linear Regression otherwise non – linear regression.

There are two lines of Regression

(i) Line of Regression of y on x

(ii) Line of Regression of x on y

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DPP- 10

Topic : Rank Correlation

Q.1 *Obtain the rank correlation co – efficient for the following data*

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

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DPP- 9

Topic : Correlation and Coefficient of Correlation

Q. 1 *Find the correlation coefficient between x and y for the following data:*

x	60	34	40	50	45	41	22	43
y	75	32	34	40	45	33	12	30

Q. 2 *Calculate the coefficient of correlation between the following ages of husband (x) and wife (y) by taking 30 and 28 as assumed mean in case of x and y respectively:*

x:	24	27	28	28	29	30	32	33	35	35	40
y:	18	20	22	25	22	28	28	30	27	30	32

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DPP- 11

Topic : *Regression Lines*

Q. 1 *From the data given, find the equation of lines of regression of x on y and y on x. Also, calculate the correlation co − efficient.*

x:	2	4	6	8	10
y:	5	7	9	8	11

Q. 2 *For two random variables, x and y with the same mean, the two regression equations are $y = ax + b$ and $x = \alpha y + \beta$. Show that $\frac{b}{\beta} = \frac{1-a}{1-\alpha}$. Find also the common mean.* (AKTU – 2017)

$\bar{x} = \bar{y} = m$

Q. 3 *The following data regarding the heights (y) and weights (x) of 100 college students are given:* (AKTU – 2018)

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Find the equation of regression line of height on weight.

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DPP- 9

Topic : Correlation and Coefficient of Correlation

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x:	24	27	28	28	29	30	32	33	35	35	40
y:	18	20	22	25	22	28	28	30	27	30	32

Q. 5:- Find the coefficient of correlation for the following table: **(UPTU – 2011)**

x:	10	14	18	22	26	30
y:	18	12	24	6	30	36

x	y	$d_x = x-22$	$d_y = y-24$	$u = \frac{x-22}{4}$	$v = \frac{y-24}{6}$	u^2	v^2	uv
10	18	−12	−6	−3	−1	9	1	3
14	12 A	−8	−12	−2	−2	4	4	4
18	$\boxed{24}$	−4	0	−1	0	1	0	0
A $\boxed{22}$	6	0	−18	0	−3	0	9	0
26	30	4	6	1	1	1	1	1
30	36	8	12	2	2	4	4	4
$\Sigma x = 120$	$\Sigma y = 126$			$\Sigma u = -3$	$\Sigma v = -3$	$\Sigma u^2 = 19$	$\Sigma v^2 = 19$	$\Sigma uv = 12$

$n = 6$

$$r_{xy} = \frac{n\Sigma uv - \Sigma u \Sigma v}{\sqrt{n\Sigma u^2 - (\Sigma u)^2}\sqrt{n\Sigma v^2 - (\Sigma v)^2}}$$

$$r_{xy} = \frac{6 \times 12 - (-3)(-3)}{\sqrt{6 \times 19 - (-3)^2}\sqrt{6 \times 19 - (-3)^2}}$$

$$r_{xy} = \frac{72 - 9}{\sqrt{105} \times \sqrt{105}} = \frac{63}{105}$$

$$\boxed{r_{xy} = 0.6}$$

Q. 4:- *A computer while calculating correlation co – efficient between two varibales X and Y from 25 pairs of observations obtained the following results :*

$n = 25, \quad \sum X = 125, \quad \sum X^2 = 650$

$\sum Y = 100, \quad \sum Y^2 = 460, \quad \sum XY = 508.$

It was, however, later discoverede at the time of checking that he had copied down two pairs as

X	Y
6	14
8	6

while the correct values were

X	Y
8	12
6	8

Obtain the correct value of correlation co – efficient

$n = 25$

Corrected values

$\sum X = 125 - 6 - 8 + 8 + 6 = 125$

$\sum Y = 100 - 14 - 6 + 12 + 8 = 100$

$\sum X^2 = 650 - 36 - 64 + 64 + 36$

$\sum X^2 = 650$

$\sum Y^2 = 460 - 196 - 36 + 144 + 64$

$\sum Y^2 = 436$

$\sum XY = 508 - 6 \times 14 - 8 \times 6 + 8 \times 12 + 6 \times 8$

$\sum XY = 520$

$$r_{xy} = \frac{n\sum XY - \sum X \sum Y}{\sqrt{n\sum X^2 - (\sum X)^2}\sqrt{n\sum Y^2 - (\sum Y)^2}}$$

$$r_{xy} = \frac{25 \times 520 - 125 \times 100}{\sqrt{25 \times 650 - (125)^2}\sqrt{25 \times 436 - (100)^2}}$$

$$\boxed{r_{xy} = 0.67}$$

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DPP- 8

Topic : CURVE FITTING

Q.1 *Given the following experimental values:*

x:	0	1	2	3
y:	2	4	10	15

Q.2 *Using method of least squares, derive the normal equations to fit the curve $y = ax^2 + bx$. Hence fit this curve to the following data:*

x	1	2	3	4	5	6	7	8
y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Q.3 *Fit the curve $pv^Y = k$ to the following data:*

$p\ (kg/cm^2)$	0.5	1	1.5	2	2.5	3
$v\ (Liters)$	1620	1000	750	620	520	460

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DPP- 7

Topic : : 1. Fitting of Parabola
2. Fitting of Exponential Curve

Q.1 *Using the method of least squares, fit second degree parabola $y = a + bx + cx^2$ for the following data:*

x:	2	3	4	5	6
y:	14	17	20	24	29

Q.2 *Using the method of least square, fit a parabolic curve of second degree to the following data:*

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

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DPP- 6

Topic : *Fitting of Straight line*

Q.1 *By the method of least squares, find the straight line that best fits the following data:*

x:	1	2	3	4	5
y:	14	27	40	55	68

Q.2 *Using the method of least square, fit a straight line to the following data:*

x	1	2	3	4	5
y	2	4	6	8	10

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DPP- 4

Topic : *Problems based on Moments, Skewness and Kurtosis*

Q.1 *The first four moments of a distribution about the value '4' of the variable are -1.5, 17, -30 and 108. State whether the distribution is leptokurtic or platykurtic.* (U.P.T.U.2014)

Q.2 *The first four moments of a distribution are 0, 2.5, 0.7 and 18.75. Test the kurtosis of the distribution.* (M.T.U.2013)

Q.3 *The first four moments of a distribution about the value 5 of the variable are 2, 20, 40 and 50. Calculate the moments about the mean and comment upon the skewness and kurtosis of the distribution.* (G.B.T.U.2011)

Q.4 *Find the measure of skewness and kurtosis on the basis of moments for the following distribution :*

x	1	3	5	7	9
f	1	4	6	4	1

Q.5 *Calculate the first four moments about the mean for the following frequency distribution and hence find the coefficient of skewness and kurtosis and comment upon the nature of the distribution.* (U.P.T.U.2014)

Class – Interval	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	6	8	17	21	15	11	2

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DPP- 4

Topic : *Skewness*

Q.1 (i) *Define skewness of a distribution.* (UPTU-2015)

(ii) *Define the coefficients of skewness.* (MTU 2012, 2013)

Q.2. *Karl Pearson's coefficient of skewness of a distribution is 0.32, its standard deviation is 6.5 and mean is 29.6. Find the mode of the distribution.*

Q.3. *The first-three central moments of a distribution are 0, 2.5, 0.7. Find the value of the moment coefficient of skewness.* (UPTU-2015)

Q.4 *In a certain distribution, the first four moments about the point 4 are – 1.5, 17, - 30 and 308. Calculate the moment coefficient of skewness.* (UPTU-2015)

DPP- 3

Topic : *Relation between μ_r and μ'_r*
Relation between μ_r and ν_r

Q.1. *The first four moments of a distribution about $x = 2$ are $1, 2.5, 5.5$ and 16. Calcutate the first four moments about the mean and about origin.*

Q.2 *In a certain distribution, the first four mements about the point $x = 4$ are $-1.5, 17, -30$ and 308. Find the moments about mean and about origin. Also, calculate β_1 and β_2.* (UPTU-2014)

Q.3. *The first four moments of a distribution about the value 5 of the variable are 2, 20, 40 and 50. Calculate mean, variance, μ_3 and μ_4.*

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DPP- 2

Topic : *Moments*

Q.1. $Calculate\ \mu_1, \mu_2, \mu_3, \mu_4, for\ the\ series : 4, 7, 10, 13, 16, 19, 22$

Q.2 *Calculate the first four moment about* $x = 15$ *and hence find the moments about the mean of the following distribution :*

x	10	11	12	13	14	15	16	17	18	19	20	21
f	9	36	75	105	116	107	88	66	45	30	18	5

Q.3 Find the first four moments about origin for the following frequency distribution:

$Marks$	0-10	10-20	20-30	30-40	40-50
$No.of\ Students$	5	10	40	20	25

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DPP- 1

Topic : *Overview of Measures of central tendency*

Q.1. Find the mean for the following individual series

x	3	6	8	10	18

Q. 2. *Find the mean of the following frequency distribution*

x	0	1	2	3	4	5	6	7
f	1	8	28	56	70	56	28	8

Q.3. *Find the mean of the following frequency distribution*

Marks	0-10	10-20	20-30	30-40	40-50
No. of Students	5	10	40	20	25

Relation between regression Coefficient and corelation Coefficient

We know that

$$b_{yx} = \frac{r\,\sigma_y}{\sigma_x} \quad \text{—— (1)}$$

$$b_{xy} = \frac{r\,\sigma_x}{\sigma_y} \quad \text{—— (2)}$$

Multiply (1) and (2)

$$b_{yx} \times b_{xy} = \frac{r\,\cancel{\sigma_y}}{\cancel{\sigma_x}} \times \frac{r\,\cancel{\sigma_x}}{\cancel{\sigma_y}}$$

$$r^2 = b_{yx} \times b_{xy}$$

$$\boxed{r = \sqrt{b_{yx} \times b_{xy}}}$$

r is the Geometric mean of b_{yx} and b_{xy}

2) *Line of Regression of x on y*

$$\boxed{x - \bar{x} = b_{xy}(y - \bar{y})}$$

$$\boxed{b_{xy} = \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2}}$$

OR

$$\boxed{b_{xy} = r\frac{\sigma x}{\sigma y}}$$

Where

b_{xy} = *Regression coefficient of x on y*

Regression line

$2x + 4y = 11$

$2x = -4y + 11$

$x = -2y + \frac{11}{2}$

$$\boxed{b_{xy} = -2}$$

Note: *Regression lines are always passes through the point* $(\bar{x}, \bar{y})$

(1) Line of Regression of y on x

$$\boxed{y - \bar{y} = b_{yx}(x - \bar{x})}$$

$$\boxed{b_{yx} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}}$$

OR

$$\boxed{b_{yx} = r\frac{\sigma y}{\sigma x}}$$

Where

b_{yx} = Regression coefficient of y on x

Equation of a line passing through a given point (x_1, y_1) and given slope (m)

$$\boxed{y - y_1 = m(x - x_1)}$$

$$2x + 4y = 11$$

$$4y = -2x + 11$$

$$y = -\frac{1}{2}x + \frac{11}{4}$$

$$m = -\frac{1}{2}$$

$$\boxed{b_{yx} = -\frac{1}{2}}$$

Note: Regression lines are always passes through the point $(\bar{x}, \bar{y})$

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DPP- 4

Topic : *Skewness*

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